
VietLedgr

VietLedgr
Use-Case Specification

Version <1.1>

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Use-Case Specification: <Use-Case Name>	Date: <13/11/2025>
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Revision History

Date	Version	Description	Author
<06/11/2025>	<1.0>	Creation of Use-Case Specification	Lê Minh Nghĩa
<13/11/2025>	<1.1>	Modification of the Use-Case Model	Lê Minh Nghĩa

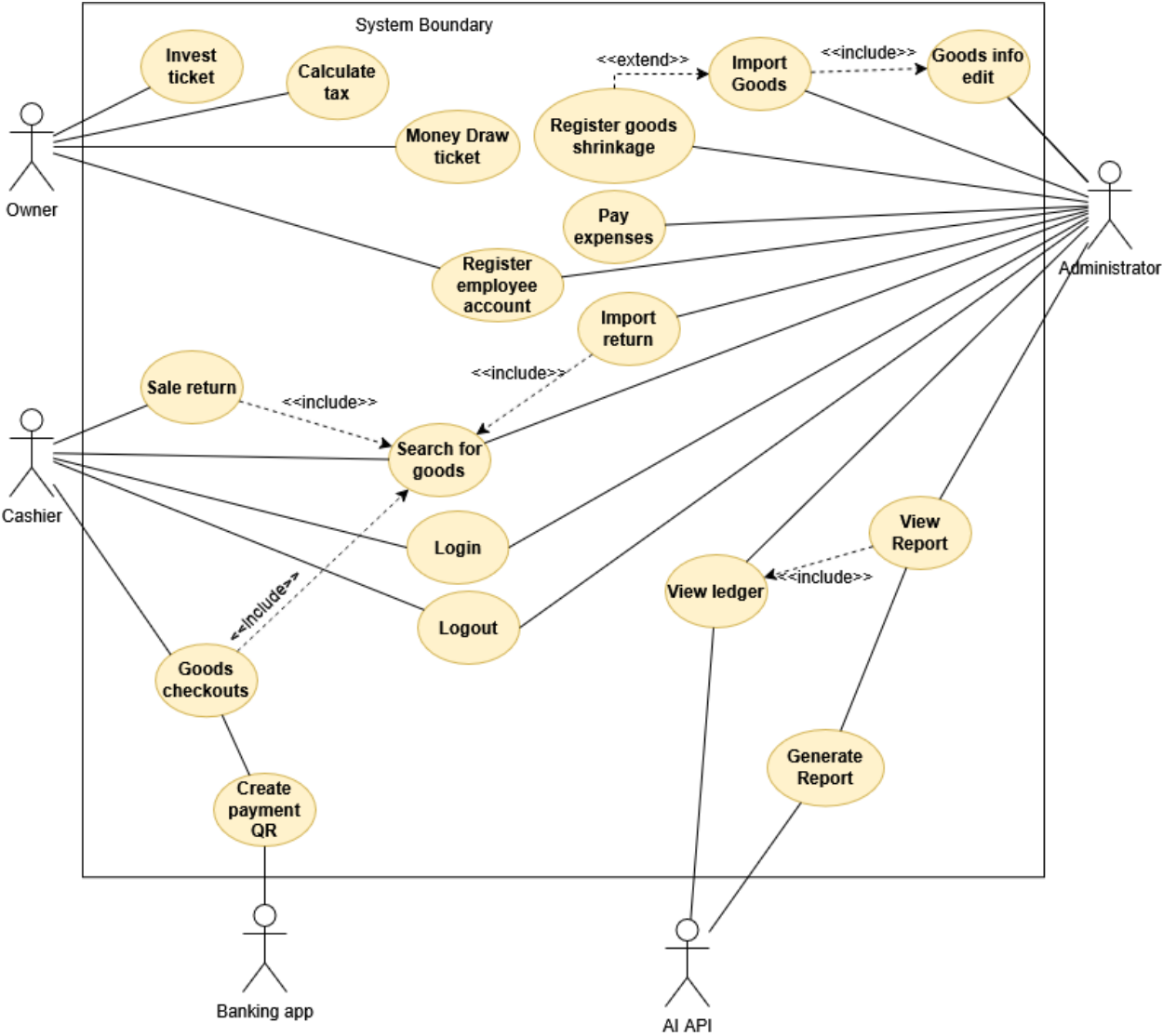
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1. Use-case Model



2. Use-case Specifications

2.1 Use-case: Import Goods

Use case Name	Import goods to the stock.
Brief description	This use case describes how the Administrator can record imported goods into the system.
Actors	Administrator
Basic Flow	- At the homepage, the user switch into the “Inventory” tab - User clicks on ‘New Item’ button - System open a menu (form) for batch input - User type in the info of the products, batch size,.... - System adds a new batch into the “Inventory” of the warehouse - System add entries into the ledger and update the accounts

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Alternative Flows	Alternative flow 1: Advanced user willing to provide more detail of the goods 1. Follow up to step #3 of the basic flow 2. User clicks on "Advance" 3. System provide detailed form for the user to input. 4. Follow step #5 and #6 of the basic flow. Alternative flow 2: Modify the item after the goods is registered 1. Follow step up to step #6 of the basic flow. 2. After the system added the item into inventory, it displays a notification. 3. Users click on "view item" 4. Users will be navigated to "Edit Goods Info" screen.
Pre-conditions	Administrator must be logged in and on the main dashboard page.
Post-conditions	The user successfully adds new item to the "inventory" and register the transaction into the ledger.

2.2 Use-case: Admin Search for Product (Dashboard)

Use case Name	Search for Product from Main Dashboard
Brief description	This use case describes how the Administrator finds a specific product from the main admin dashboard.
Actors	Administrator
Basic Flow	1. Administrator is on the main dashboard. 2. Administrator types the product's name or SKU into the global search bar. 3. Administrator presses the 'Enter' key or clicks the search icon. 4. System provides a list of Items (Batch) that fit the prompt. 5. Administrator click on an item. 6. System navigates to the "Product Details" page for that item.
Alternative Flows	Alternative flow 1: No product found 1. Follow steps 1-3 of the basic flow. 2. The system finds no product matching the query. 3. System displays a "No results found" message on the dashboard.
Pre-conditions	Administrator must be logged in and on the main dashboard page.
Post-conditions	The Administrator has successfully navigated to the desired product's detail page or is viewing a list of relevant results.

2.3 Use-case: Cashier Search for Product (POS)

Use case Name	Search for Product from POS Screen
Brief description	This use case describes how the Cashier finds a product to add to a customer's transaction.
Actors	Cashier
Basic Flow	1. At the POS screen, the Cashier clicks on the "Search Product" input field.

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	- Cashier types the product's name or SKU. - System displays a real-time list of matching products below the search bar. - Cashier clicks on the correct product from the list. - System adds the product to the current sales transaction list (cart).
Alternative Flows	Alternative flow 1: Using barcode scanner - The cursor is active in the "Search Product" input field. - Cashier scans the product's barcode. - The system automatically inputs the barcode (SKU) into the field and searches. - System finds the exact product match. - System adds the product to the current sales transaction list. Alternative flow 2: Product not found - Follow steps 1-3 of the basic flow. - The system finds no product matching the query. - System displays a "Product not found" message. Alternative flow 3: Add product into transaction list - Follow steps 1-3 of the basic flow. - Enter amount. - Click the "add to cart" button.
Pre-conditions	Cashier must be logged in and on the main POS (Point of Sale) screen with an active or new transaction.
Post-conditions	The desired product is added to the sales transaction list, or the Cashier is notified that the product is not in the system.

2.4 Use-case: User Login

Use case Name	User Login
Brief description	This use case describes how an Administrator or Cashier logs into the system.
Actors	Administrator, Cashier, Owner
Basic Flow	- User navigates to the system's login page. - User enters their email or username into the "Username" field. - User enters their password into the "Password" field. - User clicks the "Login" button. - System validates the credentials. - System logs the user in and redirects them to their respective start page (Admin Dashboard for Administrator, POS Screen for Cashier).
Alternative Flows	Alternative flow 1: Invalid credentials - Follow steps 1-4 of the basic flow. - System fails to validate the credentials (wrong username or password). - System displays an "Invalid username or password" error message. - User remains on the login page. Alternative flow 2: User forgot password - At step 1, the user clicks the "Forgot Password?" link. - This initiates the "Password Reset" use case (documented separately).

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Pre-conditions	User has a valid account and is currently not logged in. User is on the login page [website.VietLedger.com]/login.
Post-conditions	The user is successfully authenticated and has an active session in the system.

2.5 Use-case: User Logout

Use case Name	User Logout
Brief description	This use case describes how a logged-in user securely ends their session.
Actors	Administrator, Cashier, Owner
Basic Flow	1. Users click on their profile icon or name (usually in the navigation bar). 2. System displays a user menu. 3. User clicks the "Logout" button. 4. System terminates the user's session. 5. System redirects the user to the login page.
Alternative Flows	Alternative flow 1: Automatic session timeout 1. The user has been inactive for a pre-defined period (e.g., 60 minutes). 2. System automatically terminates the session for security. 3. When the user next attempts an action, the system redirects them to the login page, often with a message like "Your session has expired."
Pre-conditions	The user is currently logged into the system.
Post-conditions	The user's session is terminated, and they are no longer authenticated.

2.6 Use-case: Cashier Checkout Goods

Use case Name	Cashier Checkout Goods
Brief description	This use case describes how the Cashier finalizes a sale, accepts payment, and completes a transaction.
Actors	Cashier
Basic Flow	1. The POS screen has one or more items in the current transaction list (cart). This is achieved by the search screen. 2. Cashier clicks the "Checkout" button. 3. System displays the total amount due (including taxes). 4. Cashier asks for and selects the payment method (e.g., "Cash", "Card"). 5. Cashier enters the amount received. 6. System calculates and displays the change due. 7. Cashier confirms the transaction. 8. System records the sales, and the inventory level for the sold items is reduced. 9. System adds entries to the ledger (e.g., Debit Cash, Credit Sales Revenue; Debit COGS, Credit Inventory). 10. System generates a receipt, which the Cashier can print.

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Alternative Flows	None
Pre-conditions	Cashier is logged in at the POS screen, and there is at least one item in the cart.
Post-conditions	The sale is recorded, inventory is updated, payment is registered, and a receipt is generated.

2.7 Use-case: Cashier Receive Sale Return

Use case Name	Cashier Checkout Goods
Brief description	This use case describes how the Cashier processes a customer's return of a previously purchased item.
Actors	Cashier
Basic Flow	1. From the POS screen, Cashier navigates to the "Return" mode. 2. Cashier enters the original transaction ID from the customer's receipt. 3. System retrieves and displays the original transaction items. 4. Cashier selects the item(s) and quantity being returned. 5. Cashier clicks "Process Return". 6. System calculates the refund amount. 7. System issues the refund to the original payment method (or as store credit). 8. System updates inventory (increases stock for the returned item). 9. System adds reversal entries to the ledger (Debit Sales Returns, Credit Cash; Debit Inventory, Credit COGS). 10. System generates a return receipt.
Alternative Flows	None
Pre-conditions	Cashier is logged in at the POS screen.
Post-conditions	The return is processed, inventory is restocked, the customer is refunded, and the ledger is updated.

2.8 Use-case: Register Good Shrinkage

Use case Name	Register Good Shrinkage
Brief description	This use case describes how an Administrator records inventory losses not due to sales (damaged, expired, or stolen goods).
Actors	Administrator, Owner
Basic Flow	1. Administrator navigates to the "Inventory" tab. 2. Administrator searches for and selects the product to adjust. 3. Administrator clicks on "Record Shrinkage". 4. System displays an adjustment form. 5. Administrator enters the quantity to be removed (e.g., "-2"). 6. Administrator selects a reason (e.g., "Damaged", "Expired").

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	7. Administrator confirms the adjustment. 8. System reduces the inventory quantity for that item. 9. System adds entries to the ledger (Debit Shrinkage Expense, Credit Inventory).
Alternative Flows	None
Pre-conditions	Administrator is logged in and has permission to manage inventory.
Post-conditions	The inventory level for the item is corrected, and the loss is financially recorded in the ledger.

2.9 Use-case: Admin Return Imported Good

Use case Name	Admin Return Imported Good
Brief description	This use case describes how the admin records the return of goods to a supplier
Actors	Administrator
Basic Flow	1. Administrator navigates to the "Inventory" or "Suppliers" tab. 2. Administrator locates the original purchase order or bill associated with the goods. Can be achieved via search screen. 3. Administrator clicks "Create Purchase Return" or "Supplier Credit Note". 4. System displays a return form, pre-filled with items from the original purchase. 5. Administrator enters the quantity of the item(s) being returned. 6. Administrator confirms the return. 7. System reduces the "Inventory" quantity for the returned items. 8. System creates a credit note, reducing the amount owed to the supplier (Accounts Payable). 9. System adds entries to the ledger (Debit Accounts Payable, Credit Inventory).
Alternative Flows	None
Pre-conditions	Administrator is logged in and has permission to manage inventory.
Post-conditions	The inventory level for the item is corrected, and the loss is financially recorded in the ledger.

2.10 Use-case: View General Ledger

Use case Name	View General Ledger
Brief description	This use case describes how the admin views the detailed log of all double-entry accounting transactions.
Actors	Administrator
Basic Flow	1. Administrator navigates to the "Accounting" or "Reports" tab. 2. Administrator clicks on "General Ledger". 3. System displays a chronological list of all journal entries (showing date,

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	accounts, debit, and credit).
Alternative Flows	Alternative flow 1: Filter by date 1. Follow step 3 of the basic flow. 2. Administrator enters a "Start Date" and "End Date" in the filter fields. 3. System refreshes the list to show only entries within that date range. Alternative flow 2: Filter by account 1. Follow step 3 of the basic flow. 2. Administrator selects a specific account (ex: "Cash") from a dropdown. 3. System refreshes the list to show only entries that affect the "Cash" account.
Pre-conditions	Administrator is logged in and has accounting permissions.
Post-conditions	The Administrator has reviewed the financial transactions.

2.11 Use-case: Admin View Report

Use case Name	View Financial Report
Brief description	This use case describes how the Administrator generates and views key financial reports.
Actors	Administrator
Basic Flow	1. Administrator navigates to the "Reports" tab. 2. System displays a list of available reports ("Profit & Loss", "Balance Sheet", "Sales by Item"). 3. Administrator clicks on "Profit & Loss". 4. System prompts for a date range. 5. Administrator selects the range and clicks "Generate Report". 6. System displays the Profit & Loss statement for the selected period.
Alternative Flows	Alternative flow 1: Export report 1. Follow step 6 of the basic flow. 2. Administrator clicks the "Export" button. 3. System generates a PDF or CSV file of the report.
Pre-conditions	Administrator is logged in. There is financial data in the system.
Post-conditions	The Administrator has viewed or exported a financial report.

2.12 Use-case: Admin Pay Expense

Use case Name	Admin Pay Expense
Brief description	This use case describes how the admin records a payment for a general business expense such as rent, utilities.
Actors	Administrator
Basic Flow	1. Administrator navigates to the "Expenses" tab. 2. Administrator clicks "New Expense". 3. System displays a form. 4. Administrator enters the payment date, amount, and payee (e.g.,

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	"Power Company"). - Administrator selects the Expense Account (e.g., "Utilities Expense"). - Administrator selects the Payment Account (e.g., "Main Bank Account"). - Administrator clicks "Save". - System adds entries to the ledger (e.g., Debit Utilities Expense, Credit Main Bank Account).
Alternative Flows	none
Pre-conditions	Administrator is logged in. The Chart of Accounts (including expense accounts) is set up.
Post-conditions	The expense is recorded, and the ledger is updated.

2.13 Use-case: Admin Edit Goods (Batch) Info

Use case Name	Admin Edit Goods (Batch) Info
Brief description	This use case describes how the Admin updates information for an existing product or a specific batch.
Actors	Administrator
Basic Flow	- Administrator navigates to the "Inventory" tab. - Administrator searches for and clicks on the product to edit. - System displays the "Product Details" page. - Administrator clicks the "Edit" button. - System makes the fields (like "Name", "Sale Price", "SKU") editable. - Administrator updates the "Sale Price". - Administrator clicks "Save". - System updates the product information for all future transactions.
Alternative Flows	Alternative flow 1: Edit specific batch info - Follow up to step #3 of the basic flow. - Administrator scrolls to the "Batches" section on the Product Details page. - Administrator clicks "Edit" on a specific batch. - Administrator updates the "Expiry Date" for that batch. - Administrator clicks "Save". - System updates the info for only that specific batch.
Pre-conditions	Administrator is logged in. The product or batch exists in the system.
Post-conditions	The information for the product or batch is updated.

2.14 Use-case: Admin Register New Account for Other Roles

Use case Name	Admin Register New Account for Other Roles
Brief description	This use case describes how the Administrator creates a new user account (such as for a new Cashier).
Actors	Administrator, Owner

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Basic Flow	1. Administrator navigates to "Settings" -> "User Management". 2. Administrator clicks "Add New User". 3. System displays a user creation form. 4. Administrator enters the new user's name, email, and a temporary password. 5. Administrator selects a "Role" from a dropdown list ("Cashier"). 6. Administrator clicks "Create User". 7. System creates a new account with permissions limited to the "Cashier" role or "Admin" role.
Alternative Flows	None
Pre-conditions	Administrator is logged in and has administrative privileges.
Post-conditions	A new user account is created with the specified role.

2.15 Use-case: Owner Invest Money

Use case Name	Owner Invest Money
Brief description	This use case describes how the Owner records a new capital investment into the business.
Actors	Owner
Basic Flow	1. Owner navigates to the "Accounting" or "Banking" tab. 2. Owner clicks "Record Transaction" -> "Owner's Investment". 3. System displays a form. 4. Owner types in the info of the money invested. 5. Owner enters the date and the amount invested. 6. Owner clicks "Record". 7. System adds entries to the ledger (Debit Main Bank Account, Credit Owner's Equity).
Alternative Flows	None
Pre-conditions	Owner is logged in. The Chart of Accounts is set up.
Post-conditions	The business's cash and equity are increased, and the transaction is recorded.

2.16 Use-case: Owner Pay Tax

Use case Name	Calculate and pay tax
Brief description	This use case describes how the Owner could use the system to calculate the tax amount of the store, and records a payment made to a tax authority
Actors	Owner
Basic Flow	1. Owner navigates to the "Accounting" tab. 2. Owner clicks on "Tax detail" 3. The system navigates to a special tab for tax expense, which contain tax detail of the shop (VAT,...). 4. Owner clicks "Record Payment".

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	- System displays a form. - Owner enters the payment date, and amount. - Owner selects the correct tax account ("VAT Payable" or "Income Tax Expense"). - Owner selects the payment account ("Main Bank Account"). - Owner clicks "Save". - System adds entries to the ledger (Debit VAT Payable, Credit Main Bank Account).
Alternative Flows	None
Pre-conditions	Owner is logged in. The Chart of Accounts is set up.
Post-conditions	The tax payment is recorded, and the corresponding liability or expense account is updated.

2.17 Use-case: Owner Draw Money

Use case Name	Owner Draw Money
Brief description	This use case describes how the Owner records a withdrawal of funds from the business for personal use.
Actors	Owner
Basic Flow	- Owner navigates to the "Accounting" or "Banking" tab. - Owner clicks "Record Transaction" -> "Owner's Draw". - System displays a form. - Owner selects the bank account the money was withdrawn from ("Main Bank Account"). - Owner enters the date and the amount withdrawn. - Owner clicks "Record". - System adds entries to the ledger (Debit Owner's Draw, Credit Main Bank Account).
Alternative Flows	None
Pre-conditions	Owner is logged in. The Chart of Accounts is set up.
Post-conditions	The business's cash is reduced, and the withdrawal is recorded in the Owner's Draw equity account.

2.18 Use-case: Create Payment QR

Use case Name	Create payment QR
Brief description	The system will calculate total amount of money, and ask the Banking app to generate the QR code for the payment
Actors	Banking app
Basic Flow	- The total amount of money is sent to the Banking app API - The Banking app will return the QR code for the payment.
Alternative Flows	None
Pre-conditions	The store must link its banking account with the banking app via VietLedgr.

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Post-conditions	The system could display the QR for customers.
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2.19 Use-case: Generate Report

Use case Name	Generate Report
Brief description	The system will feed the AI the information of the store, and receive the report written by LLM
Actors	AI API (LLM)
Basic Flow	1. The system pack the ledger information and other data, and send it to the API with a prompt. 2. The AI will then generate report base on provided information. 3. The AI API then return the report to the system for the user to see.
Alternative Flows	None
Pre-conditions	Must have internet connection.
Post-conditions	The user could understand the financial information better.